

languages. In the following example, we can see that *is* and *my* have been removed from the sentence.

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
text = "India is my country"
text_tokens = word_tokenize(text)
tokens_without_sw = [word for word in text_tokens if not word in stopwords.words()]
print(tokens_without_sw)
```

OUTPUT
['India', 'country']

Stemming and lemmatisation:

Lemmatisation is the process of converting a word to its base form. The difference between stemming and lemmatisation is that the latter considers the context and converts the word to its meaningful base form, whereas stemming just removes the last few characters, often leading to incorrect meanings and spelling errors.

Based on the applicability, you can choose any of the following lemmatisers:

- Wordnet Lemmatiser
- Spacy Lemmatiser
- TextBlob
- CLiPS Pattern
- Stanford CoreNLP
- Gensim Lemmatiser
- TreeTagger

However, stopword removal or stemming/lemmatisation are not required in case text mining uses complex models like neural networks. For relatively simpler model input, generally, stopwords removal or stemming/lemmatisation helps to reduce the input text size. Here is one short example using Wordnet Lemmatiser:

```
nltk.download('wordnet')
import nltk
from nltk.stem import WordNetLemmatizer
wordnet_lemmatizer = WordNetLemmatizer()
text = "Playing Plays Played "
```

```
tokenization = nltk.word_tokenize(text)
for w in tokenization:
    print("Lemma for {} is {}".format(w, wordnet_lemmatizer.lemmatize(w)))
OUTPUT
Lemma for Playing is Playing
Lemma for Plays is Plays
Lemma for Played is Played
```

Parts of speech tagging: In your childhood, you would have heard the phrase 'parts of speech' (PoS). No doubt, it took us a lot of time to understand what adjectives and adverbs actually were, what was the difference between them, and so on. PoS tagging algorithms can predict the PoS of the given word with a higher degree of precision. You can get the PoS of individual words as a tuple.

In the following example of code, for the text content in a document, we can use the PoS to easily get only the proper nouns. Let's say we don't want to miss out on any country name from the text, we can do this very easily, too.

```
import nltk
from nltk.tokenize import PunktSentenceTokenizer

document = """Domestic cricket seasons in Australia, New Zealand, South Africa, India, Pakistan, Sri Lanka, Bangladesh, Zimbabwe and the West Indies may therefore span two calendar years, A cricket season in England is described as a single year. e.g. "2009". An international ODI series or tournament may be for a much shorter duration, and Cricinfo treats this issue by stating any series or matches. In the record tables, a two-year span generally indicates that the record was set within a domestic season in one of the above named countries."""
sentences = nltk.sent_
```

```
tokenize(document)

data = []
for sent in sentences:
    data = data + nltk.pos_tag(nltk.word_tokenize(sent))

for word in data:
    if 'NNP' in word[1]:
        print(word)
```

OUTPUT
('Australia', 'NNP')
('New', 'NNP')
('Zealand', 'NNP')
('South', 'NNP')
...
...
('England', 'NNP')
('ODI', 'NNP')
('Cricinfo', 'NNP')

Word cloud: 'A picture is worth a thousand words' is a popular proverb. Sometimes, it becomes cumbersome to read and conclude what a large body of text means. Instead, if you want an impressive visualisation format to highlight important textual data points, using a word cloud can make dull data sizzle and immediately convey crucial information.

A word cloud is a data visualisation technique used for representing text data, in which the size of each word indicates its frequency or importance. Significant textual data points can be highlighted using a word cloud. These are widely used for analysing data from social network websites. Word clouds are not just for researchers; marketers use them to convey customer sentiments.

Think about a huge body of text (like an extract from a news article in *The Financial Express* dated May 30, 2020) that's shown below. We need to know what it is about, so we create a word cloud of it using the code below. In a second, it is clear that the text is about the spread of coronavirus and the lockdown.